Course notes for

# Algebraic Topology II (KSM4E02)

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`long exact sequence from short exact sequence of chain complexes – relative singular homology – long exact sequence of singular homology – chain homotopy invariance – singular homology of contractible space`

### 7.1 Long Exact Sequence in Homology

So far we have been working with only Abelian groups, which are nothing but $\mathbb{Z}$-modules. More generally, we can fix a commutative ring $R$, and work with $R$-modules. Generalizing even further, we can consider any arbitrary *Abelian category* $\mathcal{A}$, where $0$-maps and taking quotients makes sense (see Definition 6.27). Then, we have the chain complex of objects from $\mathcal{A}$, denoted as $\mathrm{Ch}(\mathcal{A})$. The definition of $n^{\text{th}}$-homology group makes sense for $\mathrm{Ch}(\mathcal{A})$ as well. Homological algebra is the study of the (co)homology of (co)chain complex over some Abelian category.

**Theorem 7.1:** (*Long Exact Sequence of Homology of Chain Complex*)

Given a short exact sequence $0 \to A_\bullet \xrightarrow{\iota_\bullet} B_\bullet \xrightarrow{j_\bullet} C_\bullet \to 0$ of chain complexes, there exists a long exact sequence of homology groups :

$$\cdots \to H_{n+1}(C) \xrightarrow{\partial} H_n(A) \xrightarrow{\iota_\star} H_n(B) \xrightarrow{j_\star} H_n(C) \xrightarrow{\partial} H_{n-1}(A) \to \cdots,$$

which is natural with respect to maps of short exact sequences of chain complexes.

**Proof** : We consider part of the diagram

$$\begin{tikzcd}
 & \ker \partial_n^A \arrow[d, hook] & \ker \partial_n^B \arrow[d, hook] & \ker \partial_n^C \arrow[d, hook] & \\
0 \arrow[r] & A_n \arrow[r, "\iota_n"] \arrow[d, "\partial_n^A"'] & B_n \arrow[r, "j_n"] \arrow[d, "\partial_n^B"'] & C_n \arrow[r] \arrow[d, "\partial_n^C"'] & 0 \\
0 \arrow[r] & A_{n-1} \arrow[r, "\iota_{n-1}"] \arrow[d, two heads] & B_{n-1} \arrow[r, "j_{n-1}"] \arrow[d, two heads] & C_{n-1} \arrow[r] \arrow[d, two heads] & 0 \\
 & \operatorname{coker} \partial_n^A & \operatorname{coker} \partial_n^B & \operatorname{coker} \partial_n^C &
\end{tikzcd}$$

Since both the rows are exact, applying the snake lemma (Lemma 6.29), we have the exact sequence

$$0 \to \ker \partial_n^A \to \ker \partial_n^B \to \ker \partial_n^C \to \operatorname{coker} \partial_n^A \to \operatorname{coker} \partial_n^B \to \operatorname{coker} \partial_n^C \to 0.$$

This holds for each $n$. In particular, we consider the following diagram

$$\begin{tikzcd}
 & \operatorname{coker} \partial^A_{n+1} \arrow[r] \arrow[d, "\delta^A_n"'] & \operatorname{coker} \partial^B_{n+1} \arrow[r] \arrow[d, "\delta^B_n"'] & \operatorname{coker} \partial^C_{n+1} \arrow[r] \arrow[d, "\delta^C_n"'] & 0 \\
0 \arrow[r] & \ker \partial^A_{n-1} \arrow[r] & \ker \partial^B_{n-1} \arrow[r] & \ker \partial^C_{n-1} &
\end{tikzcd}$$

By above, the rows are exact. The maps $\delta^A_n, \delta^B_n, \delta^C_n$ are induced from $\partial^A_n, \partial^B_n, \partial^C_n$ respectively. Thus, the diagram is commutative. Observe that $\ker \delta^A_n = H_n(A)$ and $\operatorname{coker} \delta^A_n = H_{n-1}(A)$, and similarly for $\delta^B_n, \delta^C_n$. Indeed, we have the diagram which might be of help

$$\begin{tikzcd}
A_{n+1} \arrow[r, "\partial^A_{n+1}"] & A_n \arrow[r, "\partial^A_n"] \arrow[d, two heads] & A_{n-1} \arrow[r, "\partial^A_{n-1}"] & A_{n-2} \\
H_n(A) \arrow[r, hook] & \operatorname{coker} \partial^A_{n+1} \arrow[r, "\delta^A_n"'] \arrow[ur, "\widetilde{\partial^A_n}"] & \ker \partial^A_{n-1} \arrow[r, two heads] \arrow[u, hook] & H_{n-1}(A)
\end{tikzcd}$$

Hence, again applying the snake lemma, we have an exact sequence

$$H_n(A) \to H_n(B) \to H_n(C) \xrightarrow{\partial} H_{n-1}(A) \to H_{n-1}(B) \to H_{n-1}(C).$$

Chasing the diagram carefully, shows that the connecting maps are induced by $\iota_\bullet$ and $j_\bullet$. Pasting these exact sequences together, we get the long exact sequence in homology. Since snake lemma produces *natural* exact sequences, again a diagram chase shows that the long exact sequence is natural. $\square$

**Remark 7.2:** (*The Boundary Map*)

Let us describe the boundary map explicitly. Consider part of the diagram

$$\begin{tikzcd}
 & B_n \arrow[r, two heads, "j_n"] \arrow[d, "\partial^B_n"'] & C_n \\
A_{n-1} \arrow[r, hook, "\iota_n"] & B_{n-1} & \\
 & z \arrow[r, mapsto] & \partial^B_n(y) \arrow[r, mapsto] & 0
\end{tikzcd}
\qquad
\begin{tikzcd}
y \arrow[r, mapsto] \arrow[d, mapsto] & x \arrow[d, mapsto] \\
\partial^B_n(y) \arrow[r, mapsto] & 0
\end{tikzcd}$$

Suppose $\alpha \in H_n(C)$ is represented as $\alpha = [x]$ for some $x \in C_n$, with $\partial(x) = 0$. Then, it follows that $\partial(\alpha) = [z]$. The well-definedness is a consequence of the snake lemma. Thus, we can 'define' $\partial([x]) = [\iota_n^{-1} \partial^B_n j_n^{-1}(x)]$.

## 7.2 Relative Singular Homology and Long Exact Sequence

Let us now extend our definition of singular homology to pairs of spaces $(X, A)$ with $A \subset X$. Given such a pair $(X, A)$, observe that any singular $n$-simplex $\sigma : \Delta^n \to A$ is naturally an $n$-simplex in $X$ via the inclusion map $\iota : A \hookrightarrow X$. Thus, we have an *injective* map $S_\bullet(\iota) : S_\bullet(A) \hookrightarrow S_\bullet(X)$. Let us define the cokernel

$$S_n(X, A) := \frac{S_n(X)}{S_n(A)}.$$

Observe that $S_n(X, A)$ is an Abelian group freely generated by singular $n$-simplexes $\sigma : \Delta^n \to X$ such that $\operatorname{im}(\sigma)$ is not *completely* contained in $A$. We have the well-defined *relative* boundary map

$$\begin{aligned}\partial_n^{X,A} : S_n(X,A) &\longrightarrow S_{n-1}(X,A)\\ \left[\sum a_\sigma \sigma\right] &\longmapsto \sum a_\sigma [\partial_n^X(\sigma)].\end{aligned}$$

In particular, if we have an $n$-chain in $X$ whose boundary is compleletly contained in $A$, then its relative boundary is zero. It follows that $S_\bullet(X,A)$ is a chain complex, which is called the *relative singular chain complex*.

**Exercise 7.3:** (*Relative Singular Chain Complex is Functorial*)

Verify that $S_\bullet : \mathrm{TopPair} \to \mathrm{Ch}$ is a functor, and we can identify $S_n(X) = S_n(X,\emptyset)$ (which justifies the same notation). Moreover, $j : X = (X,\emptyset) \to (X,A)$ induces the quotient map $S_\bullet(X) \to S_\bullet(X,A)$.

The homology groups

$$H_n(X,A) := H_n(S_n(X,A))$$

are called the *relative singular homology groups* of the pair $(X,A)$. By Exercise 6.7 and Exercise 7.3, it follows that the relative singular homology groups are functors as well. In particular, given $f : (X,A) \to (Y,B)$, we have $H_n(f) = H_n(S_\bullet(f))$.

We have a short exact sequence of chain complexes

$$0 \to S_\bullet(A) \overset{\iota_\bullet}{\hookrightarrow} S_\bullet(X) \overset{j_\bullet}{\twoheadrightarrow} S_\bullet(X,A) \to 0,$$

where $\iota : A \hookrightarrow X$ and $j : X \hookrightarrow (X,A)$ are the space level maps.

**Theorem 7.4:** (*Long Exact Sequence of Singular Homology*)

There exists a natural long exact sequence

$$\cdots \to H_n(A) \overset{\iota_\star}{\to} H_n(X) \overset{j_\star}{\to} H_n(X,A) \overset{\partial}{\to} H_{n-1}(A) \to \cdots \to H_0(A) \overset{\iota_\star}{\to} H_0(X) \overset{j_\star}{\to} H_0(X,A) \to 0.$$

**Proof :** The proof is immediate from Theorem 7.1. □

Theorem 7.4 justifies that singular homology satisfies the exactness axiom.

## 7.3 Chain Homotopy Invariance

Let us recall the notion of *chain maps* $f_\bullet : C_\bullet \to D_\bullet$ (Definition 1.13), which is given a by a collection of maps $f_n : C_n \to D_n$ such that $f_{n-1} \circ \partial_n^C = \partial_n^D \circ f_n$ holds.

Given a chain complex $C_\bullet$, the *shift (or translation)* of $C_\bullet$ by degree $k$ is the chain complex $C[k]_\bullet$ defined as $C[k]_n = C_{n+k}$ and $\partial_n^{C[k]} = (-1)^k \partial_{n+k}^C$. We shall see later why the sign $(-1)^k$ appears in the definition. A *degree $k$ chain map* $C_\bullet \to D_\bullet$ is a chain map $C_\bullet \to D[k]_\bullet$. In particular, a chain map $C_\bullet \to D_\bullet$ has degree $0$.

**Exercise 7.5:** (*Shift Functor*)

Let $\Sigma^k : \mathrm{Ch} \to \mathrm{Ch}$ be the functor that shifts a chain complex by degree $k$. Check that $\Sigma^k \circ \Sigma^l = \Sigma^{k+l}$ and $\Sigma^0 = \mathrm{Id}$. In particular, $\Sigma^k$ is an *isomorphism* with inverse $\Sigma^{-k}$.

**Definition 7.6:** (*Chain Map and Chain Homotopy*)

Given two chain maps $f_\bullet, g_\bullet : C_\bullet \to D_\bullet$, a *chain homotopy* between them is a degree $1$ map $h_\bullet : C_\bullet \to D_{\bullet+1}$ such that $f - g = \partial \circ h + h \circ \partial$ holds. That is, we have the following diagram

$$\begin{tikzcd}
\cdots \arrow[r] & C_{n+1} \arrow[r, "\partial^C_{n+1}"] \arrow[d, shift right, "f_{n+1}"'] \arrow[d, shift left, "g_{n+1}"] & C_n \arrow[r, "\partial^C_n"] \arrow[d, shift right, "f_n"'] \arrow[d, shift left, "g_n"] \arrow[dl, "h_n"'] & C_{n-1} \arrow[r] \arrow[d, shift right, "f_{n-1}"'] \arrow[d, shift left, "g_{n-1}"] \arrow[dl, "h_{n-1}"] & \cdots \\
\cdots \arrow[r] & D_{n+1} \arrow[r, "\partial^D_{n+1}"'] & D_n \arrow[r, "\partial^D_n"'] & D_{n-1} \arrow[r] & \cdots
\end{tikzcd}$$

where, we have $f_n - g_n = h_{n-1} \circ \partial^C_n + \partial^D_{n+1} \circ h_n$

**Exercise 7.7:** (*Chain Homotopy is an Equivalence Relation*)

Check that the chain homotopy is an equivalence relation on the collection of all chain maps $C_\bullet \to D_\bullet$.

**Proposition 7.8:** (*Homology of Chain Homotopic Maps*)

Suppose $f_\bullet, g_\bullet : C_\bullet \to D_\bullet$ are homotopic chain maps. Then, they induce the same map between homology groups.

**Proof :** Suppose $f - g = \partial h + h\partial$ for a chain homotopy $h : C_\bullet \to D_{\bullet+1}$. Since homology is an additive functor (Exercise 6.7), we only need to show that $H_n(\partial h + h\partial) = 0$. For a class $[\alpha] \in H_n(C)$, we have $\partial(\alpha) = 0$. Recall, $[\alpha] = \alpha + \operatorname{im} \partial$ Then, it follows

$$H_n(\partial h + h\partial)[\alpha] = [(\partial h + h\partial)(\alpha)] = [\partial h(\alpha)] = 0.$$

Thus, $H_n(f) = H_n(g)$ follows for all $n$. $\square$

## 7.4 Singular Homology of Contractible Space

Before proving the homotopy invariance for singular homology, let us start with a contractible space, say, $X$. Fix a point $x_0 \in X$. Let us define a chain map $\varepsilon_\bullet : S_\bullet(X) \to S_\bullet(X)$ by setting $\varepsilon_n = 0$ for $n \neq 0$, and

$$\varepsilon_0\Big(\sum a_\sigma \sigma\Big) = \Big(\sum a_\sigma\Big)\sigma_0,$$

where $\sigma_0 : \Delta^0 \to X$ is the $0$-simplex that maps to $x_0$.

**Proposition 7.9:**

$\varepsilon_\bullet : S_\bullet(X) \to S_\bullet(X)$ is a chain map.

**Proof :** The only nontrivial part is the commutativity of the diagram

$$\begin{tikzcd} S_1(X) \arrow[r, "d_1"] \arrow[d, "\varepsilon_1 = 0"'] & S_0(X) \arrow[d, "\varepsilon_0"] \\ S_1(X) \arrow[r, "d_1"'] & S_0(X) \end{tikzcd}$$

Suppose $\sigma : \Delta^1 \to X$ is a singular 1-simplex. Then,

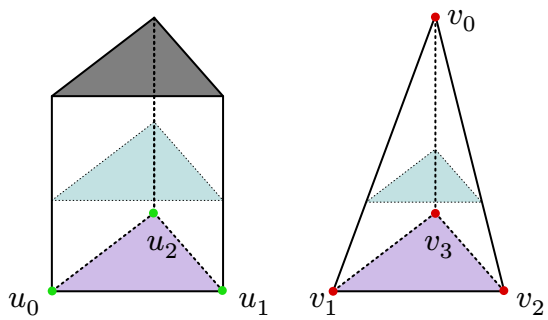
$$\varepsilon_0 d_1(\sigma) = \varepsilon_0(\sigma \circ d_0 - \sigma \circ d_1) = (1-1)\sigma_0 = 0.$$

Thus, $\varepsilon_0 d_1 = 0 = d_1 0$. This proves that $\varepsilon_\bullet$ is a chain map. $\square$

Now, suppose $h : X \times [0,1] \to X$ is a hometopy from $\mathrm{Id}_X$ to the constant map $c_{x_0} : X \to X$ which maps everything to a point $x_0 \in X$. Using $h$, we define a chain homotopy between $\varepsilon_\bullet$ and the identity chain map. Firstly, consider the map

$$\begin{aligned} q_n : \Delta^n \times [0,1] &\longrightarrow \Delta^{n+1} \\ ((t_0, ..., t_n), s) &\longmapsto (s, (1-s)t_0, ..., (1-s)t_n) \end{aligned}$$

Since $s + \sum_{i=0}^{n}(1-s)t_i = s + (1-s) = 1$, the map is well-defined, and clearly, it is continuous. As $q_n$ is a surjective map between compact $T^2$ spaces, we have $q_n$ is a quotient map.




The map $q_2 : \Delta^2 \times [0,1] \to \Delta^3$

In fact, it follows that $q_n$ identifies $\Delta^n \times \{1\}$ to the vertex $v_0 = (1, 0, ..., 0) \in \Delta^{n+1}$, and keeps every other point as it is. Note that $q_0$ is the identity map $[0,1] \to [0,1]$. Now, given a singular $n$-simplex $\sigma : \Delta^n \to X$, consider the diagram

$$\begin{tikzcd} \Delta^n \times [0,1] \arrow[r, "\sigma \times \mathrm{Id}"] \arrow[d, "q_n"] & X \times [0,1] \arrow[r, "h"] & X \\ \Delta^{n+1} \arrow[urr, dashed, "s_n(\sigma)"'] & & \end{tikzcd}$$

Since $h(X \times \{1\}) = \{x_0\}$, it follows from the universal property of a quotient map that there exists a unique continuous map $s_n(\sigma) : \Delta^{n+1} \to X$ such that

$$s_n(\sigma) \circ q_n = h \circ \left(\sigma \times \mathrm{Id}_{[0,1]}\right).$$

Extending linearly, we have a map $s_n : S_n(X) \to S_{n+1}(X)$.

Now, observe that for $0 < i \leq n+1$ we have the diagram

$$\begin{tikzcd}
\Delta^{n-1} \times [0,1] \arrow[r, "d_{i-1}^n \times \mathrm{Id}"] \arrow[d, "q_{n-1}"'] & \Delta^n \times [0,1] \arrow[r, "\sigma \times \mathrm{Id}"] \arrow[d, "q_n"] & X \times [0,1] \arrow[r, "h"] & X \\
\Delta^n \arrow[r, "d_i^{n+1}"'] \arrow[urrr, dashed, bend left, "s_{n-1}(\sigma \circ d_{i-1}^n)"] & \Delta^{n+1} \arrow[urr, dashed, "s_n(\sigma)"'] & &
\end{tikzcd}$$

Hence, $s_n(\sigma)d_i^{n+1} = s_{n-1}(\sigma d_{i-1}^n)$ holds for $0 < i \leq n+1$. Clearly, $s_n(\sigma)d_0^{n+1} = \sigma$.

**Proposition 7.10:**

$s_\bullet : S_\bullet(X) \to S_{\bullet+1}(X)$ is a chain homotopy betwen $\mathrm{Id}$ and $\varepsilon_\bullet$.

**Proof :** For a 0-simplex $\sigma : \Delta^0 \to X$, we have

$$\partial(s\sigma) = (s\sigma)d_0 - (s\sigma)d_1 = \sigma - \sigma_0,$$

since $h(\sigma(0), 1) = x_0$. For a singular $n$-simplex $\sigma : \Delta^n \to X$ with $n \geq 1$, we compute

$$\begin{aligned}
\partial(s\sigma) &= \sum_{i=0}^{n+1} (-1)^i (s\sigma)d_i^{n+1} = (s\sigma)d_0^{n+1} - \sum_{i=1}^{n+1} (-1)^{i-1}(s\sigma)d_i^{n+1} \\
&= \sigma - \sum_{i=1}^{n+1} (-1)^{i-1} s(\sigma d_{i-1}^n) \\
&= \sigma - s\left(\sum_{i=0}^{n} (-1)^i \sigma d_i^n\right), \text{by replacing } i \leftrightarrow i-1, \text{and by linearity of } s \\
&= \sigma - s(\partial\sigma) \\
\Rightarrow \sigma - \underbrace{\varepsilon_n(\sigma)}_{0} &= \partial(s\sigma) + s(\partial\sigma).
\end{aligned}$$

Thus, we have $\mathrm{Id} - \varepsilon = \partial s + s\partial$. In other words, $s_\bullet$ is a chain homotopy between $\mathrm{Id}$ and $\varepsilon_\bullet$ □

**Theorem 7.11:** (*Singular Homology of Contractible Space*)

Let $X$ be a contracitble space. Then, $H_n(X) = 0$ for all $n \neq 0$.

**Proof :** Suppose $h : X \times [0,1] \to X$ is a homotopy that takes $\mathrm{Id}_X$ to the constant map $c_{x_0}$ for some point $x_0 \in X$. Then, as above, construct the chain map $\varepsilon_\bullet : S_\bullet(X) \to S_\bullet(X)$. Since by Proposition 7.10 $\mathrm{Id}$ is chain homotopic to $\varepsilon$, an application of Proposition 7.8 it follows that they give the same map in homology. But homology is an additive functor. Hence, $\mathrm{Id} = H_n(\mathrm{Id}) = H_n(\varepsilon) = 0$ for $n \neq 0$. The only way identity map can be the zero map if the object is zero. Thus, $H_n(X) = 0$ for $n \neq 0$. □

**Exercise 7.12:** ($0^{\text{th}}$-*Homology and Path Component*)

Show that $H_0(X)$ is freely generated by the path components of $X$. In particular, $H_0(X) = \mathbb{Z}$ for a path connected space.